

Bingzheng Chen

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EDUCATION

Master

Year: 2025~2028

Northwestern Polytechnical University (Major: Information and Communication Engineering)

Supervised by Yixin Yang, Vice President of NWPU

Currently Working on: DL Enhanced Direction-of-Arrival (DOA) Estimation and Source Localization (SL)

Bachelor

Year: 2021~2025 (GPA: 3.908 / 4.1)

Northwestern Polytechnical University (Major: Information Engineering)

Outstanding Bachelor Thesis of NWPU(2025), Supervised by Assoc. Prof Zhaohui Du

Excellent Graduate Student of NWPU (2025)

China National Scholarships (2024, 2023)

Outstanding Undergraduate Scholarship of NWPU (2024, 2023)

RESEARCH PROJECT

- Underwater target localization and recognition based on super-directivity beamforming (partly supported by National Natural Science Foundation of China, starts from 2024)
- Marine noise modeling and forecasting using temporal neural networks (Key Research and Development Projects of Shaanxi, starts from 2024)
- Fault detection of shipboard propeller systems (partly supported by National Natural Science Foundation of China, starts from 2022)

PUBLICATIONS

Published

- Zhaohui Du*, **Bingzheng Chen**, Xuchen Wang, Yinan Zhu, "Nested U-shape network for Robust Direction-of-Arrival Estimation", in IEEE Transactions on Vehicular Technology, vol. 74, no. 7, pp. 11081-11094, July 2025, doi: 10.1109/TVT.2025.3540006.
- **Bingzheng Chen**, Zhaohui Du, Yixin Yang*, "Snapshot-similarity-guided sparse unfolding transformer for accurate DOA estimation", Signal Processing, Vol. 246, 2026, doi: 10.1016/j.sigpro.2026.110645.

Manuscripts in Review

- **Bingzheng Chen**, Zhaohui Du, Han Zhang, Wenyan Lu, Yixin Yang, "Attention U-Net enhanced subspace methods for robust DOA estimation", submitted to IEEE Journal of Oceanic Engineering. Under Review.
- **Bingzheng Chen**, Han Zhang, Zhaohui Du, Yixin Yang, "LISTAFormer: a Self-attention Enhanced Deep-unfolding Network for Sparse DOA Estimation", submitted to IEEE Internet of Things Journal. Under Review.
- **Bingzheng Chen**, Han Zhang, Zhaohui Du, Yixin Yang, "SDD-Net: A Sparse Deep-unfolded Decoupling-denoising Network for Direction of Arrival Estimation", submitted to IEEE Transactions on Neural Networks and Learning Systems, Under Review.
- **Bingzheng Chen**, Zhaohui Du, Yixin Yang, "Deep-unfolded Transformer for Direction-of-Arrival Estimation with Small Array Aperture and Low Frequency", submitted to Ocean Engineering, Under Review.

SKILLS

Programming Skill

Python: Neural Networks (Pytorch); Data Processing and Analysis (Numpy); Scientific Calculation (Scipy, Sklearn); Matlab: Signal Processing, Filtering and Array Designing

English Levels

IELTS (7.5); CET-6 (632); CET-4 (692);